# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Computer Science and Engineering

## ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

| | | | |
|---|---|---|---|
| **Course Code:** | CSA06 | **Course Title:** | Design and Analysis of Algorithms |
| **CO Assessed:** | CO2 – Manipulation (BL1, BL2, BL3) | **Assessment:** | Assessment Tool 2 – Scenario Based Assessment |
| **Weightage:** | 20% of CIA | **Total Marks:** | 100 |
| **CO2 Statement:** | *Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)* | | |
| **Student Name:** | **B.MANU JASWANTH** | **Reg. No.:** | **192525400** |
| **Section / Batch:** | **2025** | **Date:** | **28/07/2023** |

**Instructions to Students**

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation. • Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach. • Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

| Question Type | Focus Area | Questions |
|---|---|---|
| **Scenario Based Assessment** | Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems | Q1 |

## SECTION A – SCENARIO BASED ASSESSMENT

**Code of Conduct**

*I B.MANU JASWANTH/192525400 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: MANU

## RUBRICS FOR CALCULATION

| Criteria | Wt. | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| Problem Understanding & Context Analysis | 25% | Thoroughly understands the scenario with accurate interpretation of all requirements | Clearly understands the scenario with minor gaps | Basic understanding; some aspects of the scenario unclear | Poor understanding or misinterpretation of the scenario |
| Application of Concepts | 30% | Applies appropriate concepts effectively to solve complex real world problems | Applies relevant concepts correctly with minor errors | Applies basic concepts but with limited accuracy | Fails to apply appropriate concepts |
| Analytical & Decision Making Skills | 20% | Demonstrates strong analysis and selects optimal solutions with justification | Good analysis with reasonable solutions | Limited analysis; solutions lack justification | Poor or no analysis; incorrect decisions |
| Solution Design & Approach | 15% | Well-structured, innovative, and feasible solution approach | Logical and structured solution with minor gaps | Basic solution with limited structure or feasibility | Unclear or incorrect solution approach |
| Clarity, Justification & Presentation | 10% | Clear, well-justified explanation with strong reasoning | Clear explanation with some justification | Limited clarity and weak justification | Poorly explained with no justification |

**Marks Evaluation:**

| Criteria | Wt. | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Problem Understanding & Context Analysis | 25% | | | | |
| Application of Concepts | 30% | | | | |
| Analytical & Decision-Making Skills | 20% | | | | |
| Solution Design & Approach | 15% | | | | |
| Clarity, Justification & Presentation | 10% | | | | |
| **Total Marks (100):** | | | | | |

| Faculty In-charge | Course Coordinator | Head of Department |
|---|---|---|
| Name: | Name: | Name: |
| Signature: _____ | Signature: _____ | Signature: _____ |
| Date: _____ | Date: _____ | Date: _____ |

**Q1 (a): Digital Library – Linear Search**

**a) Explain how Linear Search works in this scenario**

A digital library stores book records in an **unsorted** order because new books are frequently added, updated, or

removed. Since the records are not sorted, **Linear Search (Sequential Search)** is the most suitable searching technique.

**Working Process**

1. Start from the first book record.
2. Compare the search key (Book ID, Title, or ISBN) with the current record.
3. If the record matches, stop the search and return the book details.
4. Otherwise, move to the next record.
5. Repeat until the book is found or all records have been checked.

**Example**

Book IDs:
[B105, B210, B315, B420, B525]
Search for **B420**

| Step | Record Compared | Result |
|---|---|---|
| 1 | B105 | Not Found |
| 2 | B210 | Not Found |
| 3 | B315 | Not Found |
| 4 | B420 | Found |

**Number of Comparisons = 4**

**Q1 (b): Analyze Best, Average, and Worst Case Performance**

| Case | Situation | Comparisons | Time Complexity |
|---|---|---|---|
| Best Case | Target is the first record | 1 | **O(1)** |
| Average Case | Target is near the middle | n/2 | **O(n)** |
| Worst Case | Target is last or absent | n | **O(n)** |

**Analysis**

- **Best Case:** The required book is located immediately.
- **Average Case:** About half the records are checked before finding the book.
- **Worst Case:** Every record must be examined.

Thus, Linear Search has a **linear growth in execution time** as the number of records increases.

**Q1 (c): Justify Whether Sorting Would Improve Efficiency**

Sorting the records would improve searching efficiency because faster algorithms such as **Binary Search** could then be used.

**Advantages of Sorting**

- Binary Search reduces search time from **O(n)** to **O(log n)**.
- Suitable for libraries where searches are frequent and updates are rare.
- Faster retrieval for large databases.

**Disadvantages**

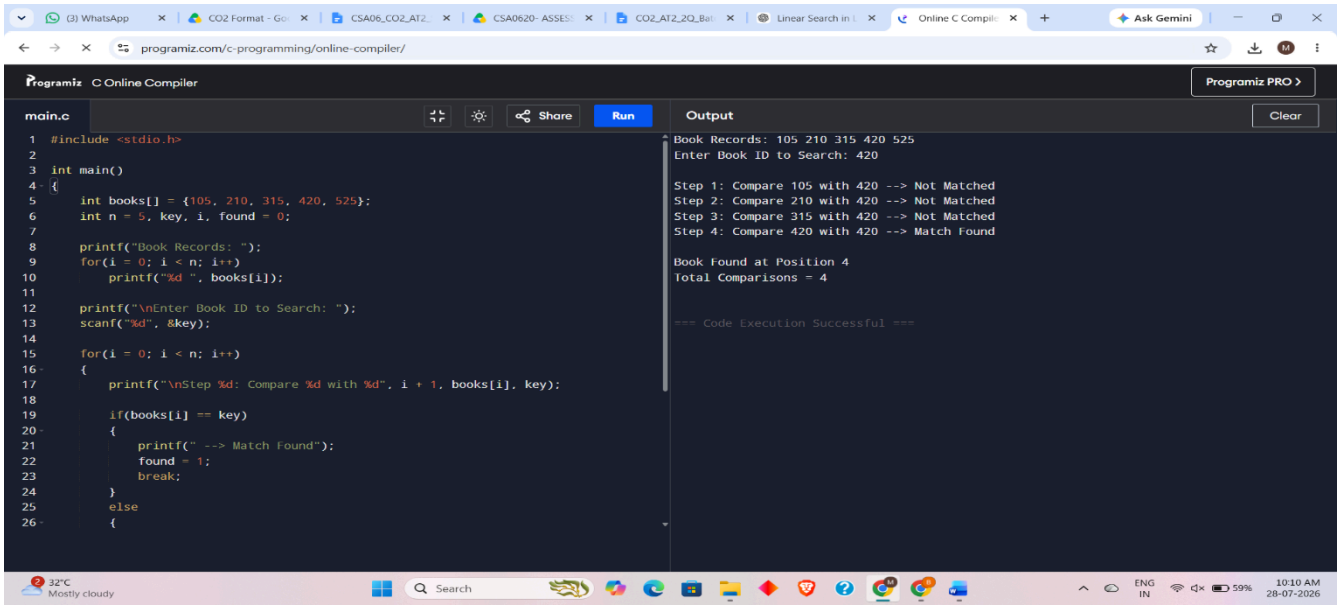
- Frequent insertion or deletion requires re-sorting.
- Maintaining sorted order increases update cost.
- Sorting itself requires additional computation.

**Justification**

Since the library is updated frequently, maintaining sorted records is expensive. Therefore, **Linear Search is more**

**practical for frequently changing data**, while **sorting is preferable only when searches greatly outnumber updates.**
**CODE:**

```
#include <stdio.h>

int main()
{
   int books[] = {105, 210, 315, 420, 525};
   int n = 5, key, i, found = 0;

   printf("Book Records: ");
   for(i = 0; i < n; i++)
      printf("%d ", books[i]);

   printf("\nEnter Book ID to Search: ");
   scanf("%d", &key);

   for(i = 0; i < n; i++)
   {
      printf("\nStep %d: Compare %d with %d", i + 1, books[i], key);

      if(books[i] == key)
      {
         printf(" --> Match Found");
         found = 1;
         break;
      }
      else
      {
         printf(" --> Not Matched");
      }
   }

   if(found)
      printf("\n\nBook Found at Position %d", i + 1);
   else
      printf("\n\nBook Not Found");

   printf("\nTotal Comparisons = %d\n", i + 1);

   return 0;
}
```


Programiz C Online Compiler
Programiz PRO
main.c
Share
Run
Output
Clear
#include <stdio.h>
int main()
{
int books[] = {105, 210, 315, 420, 525};
int n = 5, key, i, found = 0;
printf("Book Records: ");
for(i = 0; i < n; i++)
printf("%d ", books[i]);
printf("\nEnter Book ID to Search: ");
scanf("%d", &key);
for(i = 0; i < n; i++)
{
printf("\nStep %d: Compare %d with %d", i + 1, books[i], key);
if(books[i] == key)
{
printf(" --> Match Found");
found = 1;
break;
}
else
{
Book Records: 105 210 315 420 525
Enter Book ID to Search: 420
Step 1: Compare 105 with 420 --> Not Matched
Step 2: Compare 210 with 420 --> Not Matched
Step 3: Compare 315 with 420 --> Not Matched
Step 4: Compare 420 with 420 --> Match Found
Book Found at Position 4
Total Comparisons = 4
=== Code Execution Successful ===
32°C
Mostly cloudy
Search
ENG
IN
59%
10:10 AM
28-07-2026